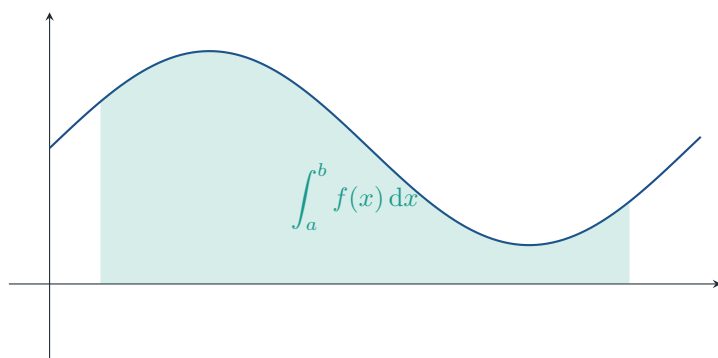

Calculus II

A Complete Study Reference

Integration Techniques · Applications · Differential Equations
Parametric & Polar · Conics · Sequences & Series



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1 Standard Integrals

These antiderivatives should be memorized; every later technique reduces a hard integral to one of these forms. (Each carries an arbitrary constant $+C$.)

Basic antiderivative table

$$\int x^n dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1)$$

$$\int \cos x dx = \sin x$$

$$\int \csc^2 x dx = -\cot x$$

$$\int \csc x \cot x dx = -\csc x$$

$$\int \sin x dx = -\cos x$$

$$\int \sec^2 x dx = \tan x$$

$$\int \sec x \tan x dx = \sec x$$

$$\int e^x dx = e^x$$

$$\int a^x dx = \frac{a^x}{\ln a}$$

$$\int \frac{1}{1+x^2} dx = \tan^{-1} x$$

$$\int \frac{1}{x} dx = \ln |x|$$

$$\int \sec x dx = \ln |\sec x + \tan x|$$

$$\int \tan x dx = \ln |\sec x| = -\ln |\cos x|$$

$$\int \csc x dx = \ln |\csc x - \cot x|$$

$$\int \cot x dx = \ln |\sin x|$$

$$\int \ln x dx = x \ln |x| - x$$

$$\int \cos^2 x dx = \frac{1}{2}x + \frac{1}{2}\sin x \cos x$$

$$\int \sin^2 x dx = \frac{1}{2}x - \frac{1}{2}\sin x \cos x$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x$$

$$\int \frac{1}{|x|\sqrt{x^2-1}} dx = \sec^{-1} x$$

Watch the signs: two table entries are commonly mis-copied. $\int \tan x dx = -\ln |\cos x|$ (a *minus* sign), and $\int \csc x dx = \ln |\csc x - \cot x|$ (a *minus* inside).

Product-to-sum identities

Indispensable for integrals of products of sines and cosines:

$$\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

2 Integration by Parts

The formula

$$\int u dv = uv - \int v du \quad \Longleftrightarrow \quad \int f(x)g'(x) dx = f(x)g(x) - \int g(x)f'(x) dx$$

It is the product rule run backwards. The whole game is choosing **which factor is u** (to differentiate) and which is dv (to integrate).

2.1 Choosing u : the LIPET rule

Pick u as the factor that comes *first* in this priority list, because differentiating it tends to simplify the problem:

L	Logarithms	$(\ln x, \log x)$
I	Inverse trig	$(\tan^{-1} x, \sin^{-1} x)$
P	Power / polynomial	$(x, x^2, \dots)$
E	Exponential	(e^x, a^x)
T	Trigonometric	$(\sin x, \cos x)$

Strategy reminders before reaching for parts: (1) try adding/splitting a term, (2) recognize $\ln x$ vs. $\tan^{-1} x$ forms, (3) use a trig identity.

Example 1 $\int \sqrt{x} \ln x \, dx$

By LIPET the logarithm wins, so $u = \ln x$ and $dv = x^{1/2} dx$:

$$u = \ln x, \quad du = \frac{1}{x} dx, \quad dv = x^{1/2} dx, \quad v = \frac{2}{3} x^{3/2}.$$

$$\begin{aligned} \int \sqrt{x} \ln x \, dx &= \frac{2}{3} x^{3/2} \ln x - \int \frac{2}{3} x^{3/2} \cdot \frac{1}{x} dx = \frac{2}{3} x^{3/2} \ln x - \frac{2}{3} \int x^{1/2} dx \\ &= \boxed{\frac{2}{3} x^{3/2} \ln x - \frac{4}{9} x^{3/2} + C} \end{aligned}$$

Example 2 $\int x \cos(4x) \, dx$

Power beats trig, so $u = x$, $dv = \cos(4x) dx$:

$$u = x, \quad du = dx, \quad dv = \cos(4x) dx, \quad v = \frac{1}{4} \sin(4x).$$

$$\int x \cos(4x) \, dx = \frac{1}{4} x \sin(4x) - \int \frac{1}{4} \sin(4x) \, dx = \boxed{\frac{1}{4} x \sin(4x) + \frac{1}{16} \cos(4x) + C}$$

2.2 Three flavours of parts

(a) Tabular method. When u is a polynomial it eventually differentiates to 0. List successive derivatives of f and successive antiderivatives of g' , then multiply along the diagonals with alternating signs $+, -, +, \dots$

Tabular example $\int 4x^2 e^{2x} \, dx$

sign	Derivatives of x^2		Antiderivatives of e^{2x}
+	x^2	$\searrow$	e^{2x}
-	$2x$	$\searrow$	$\frac{1}{2} e^{2x}$
+	2	$\searrow$	$\frac{1}{4} e^{2x}$
	0	$\searrow$	$\frac{1}{8} e^{2x}$

Multiply along each arrow with its sign, then restore the leading 4:

$$\int 4x^2 e^{2x} \, dx = 4\left(\frac{1}{2} x^2 e^{2x} - \frac{1}{2} x e^{2x} + \frac{1}{4} e^{2x}\right) + C = \boxed{2x^2 e^{2x} - 2x e^{2x} + e^{2x} + C}$$

(b) Cyclic method. When the same integral I reappears after applying parts (typically

$\int e^{ax} \sin bx \, dx$), solve for it algebraically:

$$I = (\text{boundary terms}) - I \implies 2I = (\text{boundary terms}) \implies I = \frac{1}{2}(\text{boundary terms}).$$

Cyclic example $I = \int e^x \sin x \, dx$

Parts twice (each time $u = \text{exponential}$) returns I :

$$I = e^x \sin x - e^x \cos x - I \implies 2I = e^x(\sin x - \cos x) \implies I = \frac{1}{2}e^x(\sin x - \cos x) + C.$$

(c) **With a u -substitution first**, or evaluated as a **definite integral** $[uv]_a^b - \int_a^b v \, du$. These simply combine parts with the tools of the next sections.

3 Trigonometric Integrals

A catalogue of product forms and the trick that cracks each one.

Seven standard product forms

1. $\int \sin^m x \cos^n x \, dx$ if m or n is odd, peel off one factor
2. $\int \sin^m x \sin^n x \, dx$ product-to-sum
3. $\int \cos^m x \cos^n x \, dx$ product-to-sum
4. $\int \sin(Ax) \sin(Bx) \, dx$ product-to-sum
5. $\int \cos(Ax) \cos(Bx) \, dx$ product-to-sum
6. $\int \sin(Ax) \cos(Bx) \, dx$ product-to-sum
7. $\int \sec^m x \tan^n x \, dx$ save a $\sec^2$ or a $\sec \tan$ factor

Odd-power strategy (form 1)

If **either exponent is odd**, split off a single factor so the remaining power is even, then convert with $\sin^2 x + \cos^2 x = 1$ and substitute. For example

$$\sin^3 x = \sin^2 x \cdot \sin x = (1 - \cos^2 x) \sin x,$$

and the leftover $\sin x \, dx$ becomes $-d(\cos x)$ — a clean u -substitution with $u = \cos x$.

Example $\int \sin^3 x \cos^2 x \, dx$

The sine power is odd — peel one off and let $u = \cos x$, $du = -\sin x \, dx$:

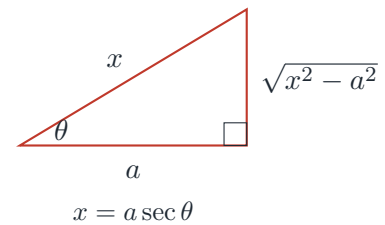
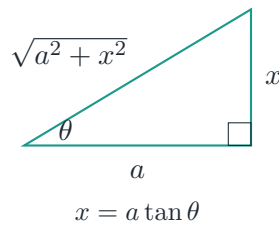
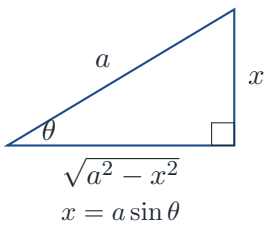
$$\begin{aligned} \int \sin^3 x \cos^2 x \, dx &= \int (1 - \cos^2 x) \cos^2 x \sin x \, dx = - \int (1 - u^2) u^2 \, du \\ &= - \int (u^2 - u^4) \, du = -\frac{1}{3}u^3 + \frac{1}{5}u^5 + C = \boxed{-\frac{1}{3}\cos^3 x + \frac{1}{5}\cos^5 x + C} \end{aligned}$$

4 Trigonometric Substitution

For integrands containing one of three quadratic radicals, substitute so the radical collapses via a Pythagorean identity.

The substitution table

Expression	Substitute	Range of θ	Identity used
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$	$\sec^2 \theta - 1 = \tan^2 \theta$



Reference triangles for back-substituting from θ to x .

5 Partial Fractions

A rational function with a factorable denominator splits into a sum of simpler fractions that integrate term-by-term.

Decomposition rules

$$\frac{C}{A \cdot B} = \frac{C_1}{A} + \frac{C_2}{B} = \frac{BC_1 + AC_2}{AB}$$

- **Distinct linear factor** $(x - r)$ contributes $\frac{A}{x - r}$.
- **Repeated linear factor** $(x - r)^2$ contributes $\frac{A}{x - r} + \frac{B}{(x - r)^2}$ — **write one term per power**.
- Always confirm the numerator degree is *less* than the denominator degree; otherwise do polynomial long division first.

For example:

$$\frac{1}{(x+1)^2(x+3)(x+2)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+3} + \frac{D}{x+2}.$$

Worked example $\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx$

Factor the denominator:

$$2x^3 + 3x^2 - 2x = x(2x^2 + 3x - 2) = x(2x - 1)(x + 2).$$

Set up the decomposition:

$$\frac{x^2 + 2x - 1}{x(2x - 1)(x + 2)} = \frac{A}{x + 2} + \frac{B}{2x - 1} + \frac{C}{x}.$$

Clearing denominators, $x^2 + 2x - 1 = Ax(2x - 1) + Bx(x + 2) + C(2x - 1)(x + 2)$. **Plug in convenient roots:**

$$x = 0 : -1 = -2C \Rightarrow C = \frac{1}{2}; \quad x = -2 : -1 = 10A \Rightarrow A = -\frac{1}{10}; \quad x = \frac{1}{2} : \frac{1}{4} = \frac{5}{4}B \Rightarrow B = \frac{1}{5}.$$

Integrate term-by-term:

$$\int \left(\frac{-1/10}{x+2} + \frac{1/5}{2x-1} + \frac{1/2}{x} \right) dx = -\frac{1}{10} \ln|x+2| + \frac{1}{10} \ln|2x-1| + \frac{1}{2} \ln|x| + C.$$

6 Improper Integrals

Integrals over an infinite interval, or with an infinite discontinuity, are defined as **limits**. If the limit is a finite number the integral **converges**; otherwise it **diverges**.

Definitions

$$\int_a^\infty f(x) dx, \quad \int_{-\infty}^a g(x) dx, \quad \int_{-\infty}^\infty h(x) dx = \int_{-\infty}^a h dx + \int_a^\infty h dx.$$

Each is computed by replacing the infinite (or singular) endpoint with a finite limit. **If there is a vertical asymptote**, set the variable equal to the asymptote value and take a one-sided limit there.

Example $\int_1^\infty \frac{1}{x} dx$

$$\int_1^\infty \frac{1}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} [\ln|x|]_1^t = \lim_{t \rightarrow \infty} (\ln t - \ln 1) = \infty.$$

The limit is infinite, so the integral **diverges**.

p -test (the benchmark family)

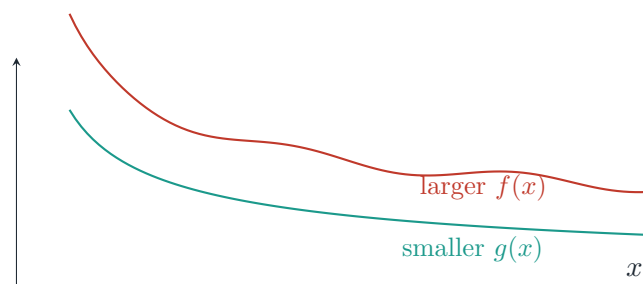
$$\int_1^\infty \frac{1}{x^p} dx \quad \begin{cases} \text{converges,} & p > 1, \\ \text{diverges,} & 0 < p \leq 1. \end{cases}$$

Comparison test

Suppose $f(x) \geq g(x) \geq 0$ on $[a, \infty)$, so that $\int_a^b f \geq \int_a^b g \geq 0$. Then

$$\int_a^\infty f \text{ converges} \Rightarrow \int_a^\infty g \text{ converges}, \quad \int_a^\infty g \text{ diverges} \Rightarrow \int_a^\infty f \text{ diverges}.$$

Idea: bound a messy integrand between simple ones. For instance compare $\int_1^\infty \frac{1 + \sin^2 x}{\sqrt{x}} dx$ against $\int_1^\infty \frac{1}{\sqrt{x}} dx$ (a divergent $p = \frac{1}{2}$ integral), forcing the original to diverge as well.



If the area under the *smaller* curve is already infinite, the larger one must be too.

7 Applications of Integration

7.1 Area between curves

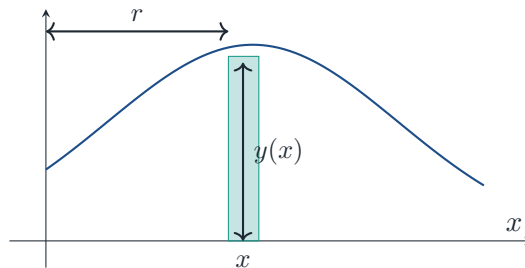
$$A = \int_a^b (\text{top} - \text{bottom}) \, dx \quad \text{or} \quad A = \int_c^d (\text{right} - \text{left}) \, dy.$$

7.2 Volume by cylindrical shells

Imagine the solid as nested thin cylinders. Integrate the **lateral surface area** of each shell ($2\pi r \cdot \text{height}$) across its thickness.

Shell formula

$$V = \int_a^b 2\pi \underbrace{r(x)}_{\text{radius}} \underbrace{h(x)}_{\text{height}} \, dx = \int_a^b 2\pi x y(x) \, dx \quad (\text{rotation about the } y\text{-axis}).$$



A single shell of radius $r = x$, height $y(x)$, and thickness dx .

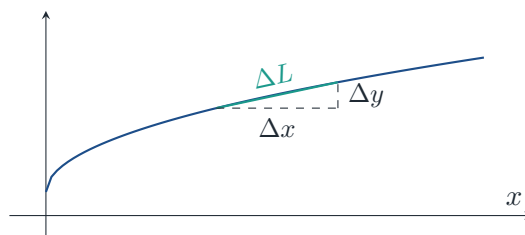
7.3 Arc length

Each tiny piece of curve is the hypotenuse of a triangle with legs Δx and Δy . Since $\Delta y = f'(x) \Delta x$,

$$\Delta L = \sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{1 + f'(x)^2} \Delta x.$$

Arc-length formula

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} \, dx.$$



$\Delta L = \sqrt{(\Delta x)^2 + (\Delta y)^2}$ — the Pythagorean origin of the formula.

7.4 Surface area of revolution

Revolve the arc; each element sweeps a band of circumference $2\pi(\text{radius})$.

Surface-of-revolution formulas

about the x -axis: $S = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx,$

about the y -axis: $S = \int_a^b 2\pi x \sqrt{1 + [f'(x)]^2} dx.$

8 Differential Equations

8.1 Separable equations

Collect each variable on its own side and integrate.

Logistic growth $\frac{dP}{dt} = 0.8 P \left(1 - \frac{P}{1000}\right)$

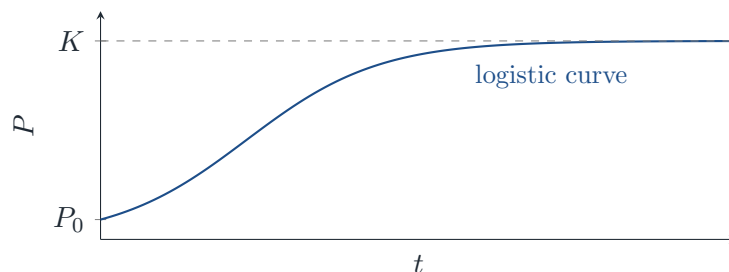
With carrying capacity $K = 1000$ and $P(0) = 100$, separate and use partial fractions:

$$\int \frac{dP}{P(1 - \frac{P}{1000})} = \int 0.8 dt, \quad \frac{1}{P(1 - P/1000)} = \frac{A}{P} + \frac{B}{1 - P/1000}.$$

Solving gives $A = 1$, $B = \frac{1}{1000}$, so integrating both sides yields

$$\ln P - \ln(1000 - P) = 0.8t + C.$$

Exponentiating and solving for P recovers the S-shaped logistic curve $P(t) = \frac{K}{1 + A e^{-kt}}$.



Population rises fastest near $P = K/2$, then levels off at the carrying capacity K .

8.2 First-order linear equations

Standard form and integrating factor

$$y' + P(x)y = Q(x).$$

Multiply through by the **integrating factor** $I(x) = e^{\int P(x) dx}$. This is engineered so the left side becomes an exact product-rule derivative:

$$\frac{d}{dx}[I(x)y] = I'(x)y + I(x)y', \quad \text{so} \quad \frac{d}{dx}[I(x)y] = I(x)Q(x).$$

Then integrate once and divide by $I(x)$.

8.3 Bernoulli equations

Reduction to linear

$$\frac{dy}{dx} + P(x)y = Q(x)y^n.$$

1. Divide by y^n : $y^{-n}\frac{dy}{dx} + P(x)y^{1-n} = Q(x)$.
2. Substitute $v = y^{1-n}$, so $\frac{dv}{dx} = (1-n)y^{-n}\frac{dy}{dx}$, giving $y^{-n}\frac{dy}{dx} = \frac{1}{1-n}\frac{dv}{dx}$.
3. Rearrange into the *linear* form $\frac{dv}{dx} + (1-n)P(x)v = (1-n)Q(x)$, and solve as above.

8.4 Two named models

Euler's method & Newton's law of cooling

Euler's method (numerical step of size h for $y' = f(x, y)$):

$$y_{n+1} = y_n + h f(x_n, y_n).$$

Newton's law of cooling (T object temperature, T_a ambient):

$$\frac{dT}{dt} = -k(T - T_a).$$

9 Limits

9.1 Four evaluation techniques

1. **Plug the value in** (when the function is continuous there): $\lim_{x \rightarrow 3} (x^2 - 10) = -1$.
2. **Factor** to cancel the offending term: $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 2$.
3. **Rationalize** by multiplying by the conjugate:

$$\lim_{x \rightarrow 0} \frac{\sqrt{3+x} - \sqrt{3}}{x} \cdot \frac{\sqrt{3+x} + \sqrt{3}}{\sqrt{3+x} + \sqrt{3}} = \lim_{x \rightarrow 0} \frac{(3+x) - 3}{x(\sqrt{3+x} + \sqrt{3})} = \frac{1}{2\sqrt{3}}.$$

4. **Divide by a clever form of 1** (e.g. by the highest power) to expose the dominant behaviour and resolve $\frac{\infty}{\infty}$ forms.

Limit laws

1. $\lim k f(x) = k \lim f(x)$
2. $\lim (f \cdot g) = \lim f \cdot \lim g$
3. $\lim \frac{f}{g} = \frac{\lim f}{\lim g}$ ($\lim g \neq 0$)
4. $\lim [f(x)]^k = [\lim f(x)]^k$
5. $\lim \log f(x) = \log [\lim f(x)]$

Famous limits

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e, \quad \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x), \quad \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = f'(a).$$

10 Parametric Equations

A curve given by $x = x(t)$, $y = y(t)$. For example $x(t) = t^2 - 2t$, $y(t) = t + 1$.

Calculus on parametric curves

Slope:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{y'(t)}{x'(t)}.$$

Arc length:

$$S = \int_a^b \sqrt{1 + (y')^2} dx = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

Area (Green's theorem):

$$A = \oint \frac{1}{2} \left(x \frac{dy}{dt} - y \frac{dx}{dt} \right) dt \quad \text{or} \quad A = \int y(t) x'(t) dt.$$

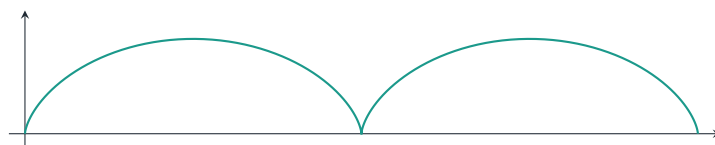
Surface of revolution:

$$(x\text{-axis}) \quad A = \int_a^b 2\pi y \sqrt{(x')^2 + (y')^2} dt, \quad (y\text{-axis}) \quad A = \int_a^b 2\pi x \sqrt{(x')^2 + (y')^2} dt.$$

The cycloid

Traced by a point on a rolling circle of radius r :

$$x = r(t - \sin t), \quad y = r(1 - \cos t).$$

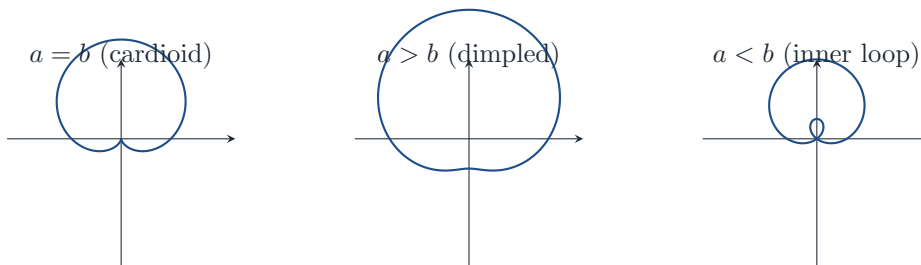


Two arches of a cycloid $x = t - \sin t$, $y = 1 - \cos t$.

11 Polar Coordinates

11.1 Limaçons and cardioids

The family $r = a + b \sin \theta$ or $r = a + b \cos \theta$ (with $a, b > 0$) changes shape according to the ratio a/b .



$r = a + b \sin \theta$ for three ratios of a to b .

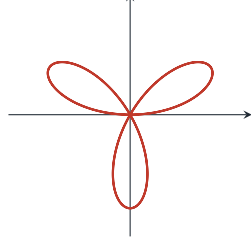
11.2 Rose curves

Petal-counting rule for $r = a \sin(n\theta)$

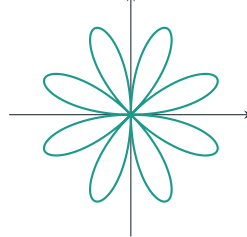
$$\begin{cases} n \text{ odd} : & n \text{ petals,} \\ n \text{ even} : & 2n \text{ petals.} \end{cases}$$

The petals each have length $|a|$. **Note:** an *odd* n gives exactly n petals (the curve retraces itself), while an *even* n doubles them to $2n$.

$r = \sin 3\theta$ (3 petals)



$r = \sin 4\theta$ (8 petals)



11.3 Polar calculus formulas

Area, arc length, tangents

Area swept by a polar curve:

$$A = \int_a^b \frac{1}{2} r^2 d\theta, \quad \text{between two curves: } A = \int_a^b \frac{1}{2} r_{\text{out}}^2 d\theta - \int_a^b \frac{1}{2} r_{\text{in}}^2 d\theta.$$

Arc length:

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta.$$

Slope of the tangent line:

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}.$$

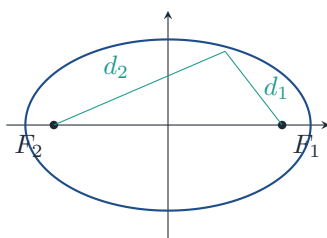
12 Conic Sections

12.1 Ellipse

Ellipse centered at (h, k)

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1, \quad a^2 = b^2 + c^2, \quad d_1 + d_2 = 2a.$$

Here a, b are the semi-axis radii, (h, k) the center, and c the focal distance. Parametrically, $x = h + a \cos \theta$, $y = k + b \sin \theta$. The defining property: the sum of distances to the two foci is constant, $d_1 + d_2 = 2a$.



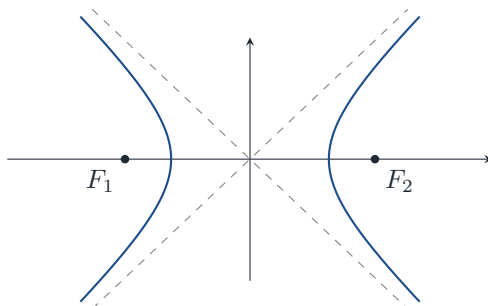
Ellipse with foci F_1, F_2 ; a point on the curve satisfies $d_1 + d_2 = 2a$.

12.2 Hyperbola

Hyperbola centered at (h, k)

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1, \quad |d_2 - d_1| = 2a.$$

The defining property is the constant *difference* of focal distances. **Note:** the constant difference is $2a$ (not a).

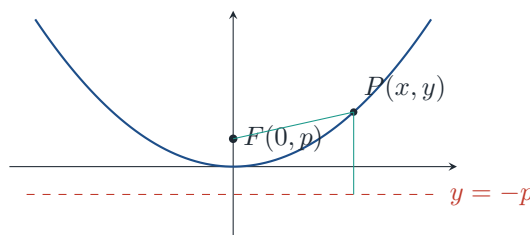


Hyperbola with its two branches and asymptotes.

12.3 Parabola

Parabola with focus and directrix

A parabola opening upward with focus $F(0, p)$ has directrix $y = -p$. Every point $P(x, y)$ is equidistant from the focus and the directrix.



Distance from P to the focus equals its distance to the directrix.

13 Sequences

A sequence is an ordered list $\{a_1, a_2, a_3, \dots, a_n\}$, also written $\{a_n\}$ or $\{a_n\}_{n=1}^{\infty}$.

Limit of a sequence

$$\lim_{n \rightarrow \infty} a_n = L.$$

If this limit exists, the sequence **converges** (approaches a single number); otherwise it **diverges**. **Infinite limit:** $\lim_{n \rightarrow \infty} a_n = \infty$ means that for every positive number M there is an

integer N such that $n > N \Rightarrow a_n > M$.

Two useful facts

From functions to sequences: if $\lim_{x \rightarrow \infty} f(x) = L$ and $f(n) = a_n$ for integer n , then $\lim_{n \rightarrow \infty} a_n = L$.

Geometric sequence: $\{r^n\}$ converges for $-1 < r \leq 1$ and diverges otherwise, with

$$\lim_{n \rightarrow \infty} r^n = \begin{cases} 0, & -1 < r < 1, \\ 1, & r = 1. \end{cases}$$

14 Power Series

Definition

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots$$

More generally, a power series **centered at** a :

$$\sum_{n=0}^{\infty} c_n (x - a)^n = c_0 + c_1 (x - a) + c_2 (x - a)^2 + \cdots$$

A power series may converge for some x and diverge for others.

Ratio test for convergence

$$L = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| \quad \Rightarrow \quad \begin{cases} L < 1 : & \text{converges,} \\ L > 1 : & \text{diverges.} \end{cases}$$

Example For which x does $\sum_{n=0}^{\infty} \frac{x^n}{(2n)!}$ converge?

With $a_n = \frac{x^n}{(2n)!}$,

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{(2(n+1))!} \cdot \frac{(2n)!}{x^n} \right| = \lim_{n \rightarrow \infty} \frac{|x|}{(2n+2)(2n+1)} = 0 < 1.$$

The limit is $0 < 1$ for *every* x , so the series converges for all real x .

14.1 Interval and radius of convergence

The three cases

For $\sum_{n=0}^{\infty} c_n (x - a)^n$ exactly one of the following holds:

1. Converges only at $x = a$ ($R = 0$).
2. Converges for all x ($R = \infty$, interval $-\infty < x < \infty$).
3. Converges if $|x - a| < R$ and diverges if $|x - a| > R$, for some positive **radius of convergence** R ; the base interval is $a - R < x < a + R$.

Finding R :

$$\text{ratio test: } L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|, \quad R = \frac{1}{L}; \quad \text{root test: } L = \lim_{n \rightarrow \infty} \sqrt[n]{|c_n|}, \quad R = \frac{1}{L}.$$

Always test the two endpoints separately — they decide whether the interval is open, half-open, or closed.

Full example $\sum_{n=0}^{\infty} \frac{n(x+2)^n}{3^{n+1}}$

Ratio test:

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)(x+2)^{n+1}}{3^{n+2}} \cdot \frac{3^{n+1}}{n(x+2)^n} \right| = \lim_{n \rightarrow \infty} \frac{n+1}{n} \cdot \frac{|x+2|}{3} = \frac{|x+2|}{3}.$$

Converges when $\frac{|x+2|}{3} < 1$, i.e. $|x+2| < 3$, giving $-5 < x < 1$.

Test the endpoints in $\sum \frac{n(x+2)^n}{3^{n+1}}$:

$$x = -5 : \frac{1}{3} \sum (-1)^n n \quad (\text{terms flip sign, } a_n \not\rightarrow 0 \Rightarrow \text{diverges});$$

$$x = 1 : \frac{1}{3} \sum n \quad (\rightarrow \infty \Rightarrow \text{diverges}).$$

Since $\lim a_n \neq 0$ at both ends, neither endpoint is included:

$$\boxed{\text{interval of convergence} = (-5, 1), \quad R = 3.}$$

14.2 Endpoint behaviour: the four interval types

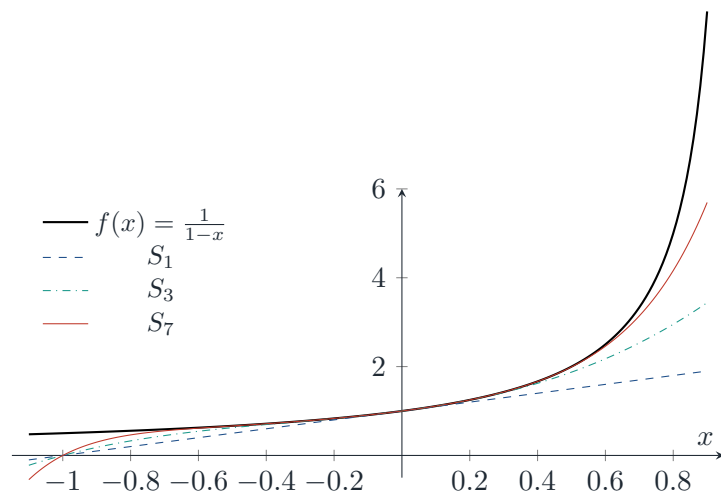
Given an interval determined by endpoints $p < q$, the center is $\frac{p+q}{2}$ and $R = \frac{q-p}{2}$; what happens at the endpoints fixes the domain.

Designing a series with a prescribed interval

Interval	Coefficient choice	Endpoint check
(p, q) open	$c_n = \frac{1}{R^n}$	geometric: diverges at both ends
$(p, q]$ half-open	$c_n = \frac{(-1)^n}{n R^n}$	conv. at q (alt. harmonic), div. at p (harmonic)
$[p, q)$ half-open	$c_n = \frac{1}{n R^n}$	div. at q (harmonic), conv. at p (alt.)
$[p, q]$ closed	$c_n = \frac{1}{n^2 R^n}$	conv. at both ($\sum 1/n^2$ & $\sum (-1)^n/n^2$)

14.3 Intuition: the geometric series $\sum x^n = \frac{1}{1-x}$

Inside $(-1, 1)$ the partial sums $S_N(x)$ hug $f(x) = \frac{1}{1-x}$ ever more tightly as N grows — that is convergence. Outside $(-1, 1)$, no matter how many terms you add the partial sums fly off to infinity. Near the endpoints $x = \pm 1$ convergence slows; at $x = 1$ the series $\sum 1$ diverges. Hence the convergent interval is exactly $(-1, 1)$.



Partial sums converge to $\frac{1}{1-x}$ on $(-1, 1)$ and diverge outside it.

End of Calculus II reference.